

opment board is a plus. Selection of the development kit nearly solidifies the choice of the microcontroller.

The last consideration is to examine the compiler and tools that are available. Most microcontrollers have a number of choices for compilers, example code and debugging tools. It is important to make sure that all the necessary tools are available for the part. Without the right tools, the development process could become tedious and expensive.

By following the above points one by one, you can arrive at the right microcontroller for your product. Once you are done with the selection, you can move ahead and select the development board for the microcontroller. For most popular microcontrollers, you can purchase the development board either from the manufacturer or a third party. However, you have to be careful while selecting a less common microcontroller because its development board might not be available. Tables I and II give overview of some popular microcontrollers from different manufacturers.

Give preference to an official board from the microcontroller manufacturer as with it you will have at least the basics right. Some small manufacturers also sell these boards but these are designed without giving much attention to the design guidelines.

Some popular development boards

The Arduino family. Arduino family

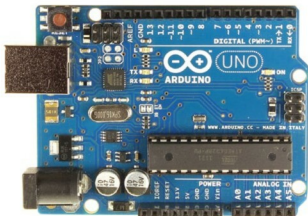


Fig. 1: Arduino UNO board

needs no introduction. It is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended for anyone interested in creating interactive objects or environments without having much programming experience. Because of the simple IDE and coding environment, Arduino is a common choice for engineers and non-engineers alike. The hardware of the various boards in the family is completely open source and designed around an 8-bit Atmel AVR microcontroller, or a 32-bit Atmel ARM. The boards come pre-burnt with a bootloader that allows you to upload new code without the use of external hardware programmer.

Raspberry Pi. Raspberry Pi has recently gained huge popularity due to its small size, great computing power and low cost. The heart of the board is a Broadcom BCM2835 system-on-a-chip (SoC) that contains a 700MHz ARM 11 chip and a GPU capable of playing movies with BluRay quality. The Raspberry Pi Foundation has allowed overclocking up to 800 MHz. The CPU power is comparable to that of an old Pentium III with at least ten times lower power consumption.

On-board hardware includes a 3.5mm audio jack, an RCA connector for connecting analogue television sets and also a modern HDMI interface for transferring digital audio and video. An RJ45 network connector is also available and a dual USB type 'A' connector allows connection for mouse and keyboards. The operating system used is LINUX, with several distributions already optimised to run on Raspberry Pi. The OS boots from the SD card. To interface with external hardware, a complete GPIO connector is provided, which has pins with different functions. The device has seven true GPIO pins, I2C interface,



Fig. 2: Raspberry Pi

SPI interface, serial T_x/R_x pins and PWM pins that can be used to control external hardware.

TI's MSP430 LaunchPad. The MSP430 LaunchPad (Part No. MSP-EXP430G2) is an easy-to-use Flash programmer and debugging tool for Texas Instruments' MSP430G2xx series microcontrollers. It has lots of debugging and interfacing features on-board for any development across MSP430

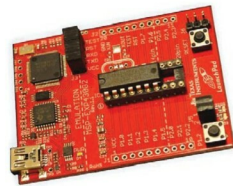


Fig. 3: MSP430 LaunchPad

microcontroller device. It has on-board emulation for programming and debugging and features a 14-/20-pin DIP socket, on-board buttons and LEDs, and BoosterPack-compatible pinouts that support a wide range of plug-in modules for added functionality such as wireless and displays.

The MSP-EXP430G2 LaunchPad also comes with two MSP430 devices, with up to 16kB Flash, 512B RAM, 16MHz CPU speed and integrated peripherals such as 8-channel 10-bit ADC, timers and serial communication (UART, I2C and SPI). Free software development tools such as TI's Eclipse-based Code Composer Studio,